

# Region and read layout determine the trimming branch

Region names guide the search; observed read starts and the library protocol decide the action.

| Assay branch          | Read layout            | Primer evidence                                               | Trim action                                               | Downstream implication                                           |
|-----------------------|------------------------|---------------------------------------------------------------|-----------------------------------------------------------|------------------------------------------------------------------|
| V4 • primer retained  | Paired short reads     | Both R1 and R2 start with the documented pair                 | Trim anchored front-f and front-r; audit paired retention | Recalculate quality and overlap; DADA2 trim-left = 0 for primers |
| V4 • primer absent    | Paired short reads     | No coherent 5′ match; protocol uses custom sequencing primers | Skip primer trimming                                      | Proceed to quality-based truncation; do not discard all reads    |
| V3-V4                 | Paired short reads     | Confirm the exact 341F/805R-family sequences and orientation  | Trim run-specific primers before denoising                | Longer insert makes overlap chemistry- and quality-dependent     |
| Full-length 16S       | Single long read       | Forward and reverse boundaries occur in one molecule          | Use linked/orientation-aware primer logic                 | No paired merge; use platform-appropriate long-read denoising    |
| Short-read single-end | One acquired direction | Probe only the sequenced 5′ end                               | Trim the documented front primer if present               | No paired merge and no evidence about the missing mate           |

Never merge runs from different primer regions before run-specific trimming and denoising.